B5 General Relativity — Model Solutions, Trinity 2023

Paper A16729W1

The rubric instructs the candidate to answer two questions out of four. Full mark-scheme solutions to all four questions are given below.

Question 1 — Weak-field gravity, gravitomagnetism, Nordström's theory

Problem. Linearised Einstein gravity, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, with weak-field Ricci tensor as quoted. Show: (a) static non-relativistic source recovers $\nabla^2\Phi = 4\pi G\rho$ with $h_{00} = -2\Phi/c^2$; (b) the test-particle equation of motion is $\ddot{\mathbf{x}} = -\nabla\Phi + \mathbf{v}\times\mathbf{B}_G$ with $\mathbf{B}_G = \nabla\times c\mathbf{h}$; (c) compare $|\mathbf{B}_G|$-acceleration to Newtonian gravity at geosynchronous orbit; (d) for Nordström's theory $g_{\mu\nu} = e^{2\Phi/c^2}\eta_{\mu\nu}$ with $R = -\kappa_N g_{\mu\nu}T^{\mu\nu}$, recover $\nabla^2\Phi = 4\pi G\rho$ and find κ_N/κ_E; (e) explain why light is undeflected and $\mathbf{B}_G = 0$.

(a) Newtonian limit

Use signature $(-,+,+,+)$. Rename the dummy index in the third term of the quoted weak-field Ricci tensor: with $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$,

$$R_{\mu\nu} = \tfrac{1}{2}\eta^{\alpha\gamma}\big(-\partial_\mu\partial_\alpha h_{\nu\gamma} + \partial_\gamma\partial_\alpha h_{\nu\mu} + \partial_\mu\partial_\nu h_{\alpha\gamma} - \partial_\nu\partial_\gamma h_{\alpha\mu}\big).$$

Specialise to $\mu = \nu = 0$ and static fields, $\partial_0 h_{\mu\nu} = 0$:

$$R_{00} = \tfrac{1}{2}\eta^{\alpha\gamma}\partial_\gamma\partial_\alpha h_{00} = \tfrac{1}{2}\nabla^2 h_{00}.$$

Trace-reverse the field equations. Take the trace $g^{\mu\nu}(R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}) = -\kappa_E T$:

$$R = \kappa_E T, \qquad R_{\mu\nu} = -\kappa_E\big(T_{\mu\nu} - \tfrac{1}{2}Tg_{\mu\nu}\big).$$

Non-relativistic source: $T_{00} = \rho c^2$, $T_{ij} = 0$:

$$T = g^{\mu\nu}T_{\mu\nu} = \eta^{00}T_{00} = -\rho c^2.$$

Hence

$$T_{00} - \tfrac{1}{2}T\eta_{00} = \rho c^2 - \tfrac{1}{2}(-\rho c^2)(-1) = \tfrac{1}{2}\rho c^2.$$

Insert into $R_{00} = -\kappa_E\big(T_{00} - \frac{1}{2}Tg_{00}\big)$:

$$\tfrac{1}{2}\nabla^2 h_{00} = -\tfrac{1}{2}\kappa_E\rho c^2.$$

Substitute $h_{00} = -2\Phi/c^2$ and $\kappa_E = 8\pi G/c^4$:

$$-\frac{1}{c^2}\nabla^2\Phi = -\frac{4\pi G}{c^4}\rho c^2.$$

Rearrange:

$$\boxed{\nabla^2\Phi = 4\pi G\rho.} \tag{1}$$

(b) Gravitomagnetic equation of motion

Geodesic equation, with τ proper time:

$$\frac{\mathrm{d}^2x^\mu}{\mathrm{d}\tau^2}+\Gamma^\mu_{\alpha\beta}u^\alpha u^\beta=0,\qquad u^\alpha=\frac{\mathrm{d}x^\alpha}{\mathrm{d}\tau}.$$

Slow motion: $u^0\approx c$, $u^i=v^i+O(v^3/c^2)$, and $\mathrm{d}\tau\approx\mathrm{d}t$:

$$\frac{\mathrm{d}^2x^i}{\mathrm{d}t^2}=-\Gamma^i_{\alpha\beta}\frac{u^\alpha u^\beta}{(u^0)^2}\,c^2\;\Rightarrow\;\frac{\mathrm{d}^2x^i}{\mathrm{d}t^2}=-c^2\Gamma^i_{00}-2c\,\Gamma^i_{0j}v^j+O(v^2).$$

Linearised Christoffels with $\partial_0 h_{\mu\nu}=0$:

$$\Gamma^i_{00}=-\tfrac12\eta^{ii}\partial_i h_{00}=\tfrac12\partial_i h_{00},\qquad \Gamma^i_{0j}=\tfrac12\eta^{ii}\big(\partial_j h_{0i}-\partial_i h_{0j}\big)=\tfrac12\big(\partial_j h_{0i}-\partial_i h_{0j}\big).$$

Static Γ^i_{00}-piece, using $h_{00}=-2\Phi/c^2$:

$$-c^2\Gamma^i_{00}=-\tfrac12c^2\partial_i h_{00}=\partial_i\Phi\Rightarrow -c^2\boldsymbol{\Gamma}_{00}=-\nabla\Phi.$$

Velocity-dependent piece:

$$-2c\,\Gamma^i_{0j}v^j=-c\big(\partial_j h_{0i}-\partial_i h_{0j}\big)v^j.$$

Define $\mathbf{h}=(h_{01},h_{02},h_{03})$ and $\mathbf{B}_G=\nabla\times c\mathbf{h}$, so $(\mathbf{B}_G)_k=c\,\epsilon_{klm}\partial_l h_{0m}$:

$$\big(\mathbf{v}\times\mathbf{B}_G\big)_i=\epsilon_{ijk}v^j(\mathbf{B}_G)_k=c\,\epsilon_{ijk}\epsilon_{klm}v^j\partial_l h_{0m}.$$

Apply $\epsilon_{ijk}\epsilon_{klm}=\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl}$ (the hint):

$$\big(\mathbf{v}\times\mathbf{B}_G\big)_i=c\,v^j\big(\partial_i h_{0j}-\partial_j h_{0i}\big).$$

Compare with the velocity-dependent piece above:

$$-2c\Gamma^i_{0j}v^j=\big(\mathbf{v}\times\mathbf{B}_G\big)_i.$$

Combine:

$$\boxed{\frac{\mathrm{d}^2\mathbf{x}}{\mathrm{d}t^2}=-\nabla\Phi+\mathbf{v}\times\mathbf{B}_G,\qquad \mathbf{B}_G=\nabla\times c\mathbf{h}.}\tag{2}$$

(c) Geosynchronous comparison

Geosynchronous radius from Kepler:

$$r_{\rm geo}=\Big(\frac{GM_\oplus T^2}{4\pi^2}\Big)^{1/3}\approx 4.22\times10^7\,\mathrm{m}.$$

Newtonian centripetal acceleration:

$$a_N=\frac{GM_\oplus}{r_{\rm geo}^2}\approx 0.224\,\mathrm{m\,s^{-2}}.$$

Orbital speed:

$$v_{\rm geo}=\sqrt{\frac{GM_\oplus}{r_{\rm geo}}}\approx 3.07\times10^3\,\mathrm{m\,s^{-1}}.$$

Earth angular momentum, with $M_\oplus = 5.97 \times 10^{24}\,\mathrm{kg}$, $R_\oplus = 6.37 \times 10^6\,\mathrm{m}$, $\omega_\oplus = 7.27 \times 10^{-5}\,\mathrm{rad\,s^{-1}}$:

$$J = \tfrac{2}{5} M_\oplus R_\oplus^2 \omega_\oplus \approx 7.05 \times 10^{33}\,\mathrm{kg\,m^2\,s^{-1}}.$$

Gravitomagnetic field amplitude:

$$B_G = \frac{2GJ}{c^2 r_{\mathrm{geo}}^3} = \frac{2 \cdot 6.67 \times 10^{-11} \cdot 7.05 \times 10^{33}}{(3 \times 10^8)^2 (4.22 \times 10^7)^3} \approx 1.4 \times 10^{-13}\,\mathrm{s^{-1}}.$$

Acceleration $a_G = v_{\mathrm{geo}} B_G$:

$$a_G \approx 3.07 \times 10^3 \cdot 1.4 \times 10^{-13} \approx 4 \times 10^{-10}\,\mathrm{m\,s^{-2}}.$$

Ratio:

$$\boxed{\frac{a_G}{a_N} \sim \frac{4 \times 10^{-10}}{0.22} \approx 2 \times 10^{-9}.}$$

The gravitomagnetic acceleration is some nine orders of magnitude smaller than Newtonian gravity at geosynchronous altitude.

(d) Nordström's scalar theory

Conformal metric, expand to first order in Φ/c^2:

$$g_{\mu\nu} = e^{2\Phi/c^2} \eta_{\mu\nu} \approx \eta_{\mu\nu} + \frac{2\Phi}{c^2} \eta_{\mu\nu} \Rightarrow h_{\mu\nu} = \frac{2\Phi}{c^2} \eta_{\mu\nu}.$$

Read off components: $h_{00} = -2\Phi/c^2$, $h_{ij} = (2\Phi/c^2)\delta_{ij}$, $h_{0i} = 0$.

Compute the linearised Ricci scalar. Trace-reverse h does not help here; use $R = \eta^{\mu\nu} R_{\mu\nu}$ with the quoted $R_{\mu\nu}$. Linearised Ricci of a pure trace $h_{\mu\nu} = \frac{2\Phi}{c^2}\eta_{\mu\nu}$:

$$R_{\mu\nu} = \tfrac{1}{2} \eta^{\alpha\gamma} \big(-\partial_\mu \partial_\alpha h_{\nu\gamma} + \partial_\gamma \partial_\alpha h_{\nu\mu} + \partial_\mu \partial_\nu h_{\alpha\gamma} - \partial_\nu \partial_\gamma h_{\alpha\mu} \big).$$

Substitute $h_{\nu\gamma} = (2\Phi/c^2)\eta_{\nu\gamma}$ and use $\eta^{\alpha\gamma}\eta_{\nu\gamma} = \delta^\alpha_\nu$:

$$R_{\mu\nu} = \frac{1}{c^2} \big(-\partial_\mu \partial_\nu \Phi + \eta_{\mu\nu} \Box \Phi + \eta^{\alpha\gamma} \eta_{\alpha\gamma} \partial_\mu \partial_\nu \Phi - \partial_\nu \partial_\mu \Phi \big).$$

Apply $\eta^{\alpha\gamma}\eta_{\alpha\gamma} = 4$:

$$R_{\mu\nu} = \frac{1}{c^2} \big(-\partial_\mu \partial_\nu \Phi + \eta_{\mu\nu} \Box \Phi + 4 \partial_\mu \partial_\nu \Phi - \partial_\mu \partial_\nu \Phi \big) = \frac{1}{c^2} \big(2 \partial_\mu \partial_\nu \Phi + \eta_{\mu\nu} \Box \Phi \big).$$

Take the trace:

$$R = \eta^{\mu\nu} R_{\mu\nu} = \frac{1}{c^2} (2 \Box \Phi + 4 \Box \Phi) = \frac{6}{c^2} \Box \Phi.$$

Static limit, $\Box\Phi \to \nabla^2 \Phi$:

$$R = \frac{6}{c^2} \nabla^2 \Phi.$$

Now the right-hand side. Trace of the energy-momentum tensor of a non-relativistic source: $g_{\mu\nu} T^{\mu\nu} = T = -\rho c^2$ (to leading order). Field equation:

$$\frac{6}{c^2} \nabla^2 \Phi = -\kappa_N (-\rho c^2) = \kappa_N \rho c^2.$$

Match to $\nabla^2 \Phi = 4\pi G \rho$:

$$\frac{\kappa_N c^4}{6} = 4\pi G \Rightarrow \kappa_N = \frac{24\pi G}{c^4}.$$

Compare with $\kappa_E = 8\pi G/c^4$:

$$\boxed{\kappa_N = 3\kappa_E.} \tag{3}$$

(e) No light deflection, $\mathbf{B}_G = 0$

Conformal metric: under $g_{\mu\nu} = e^{2\Phi/c^2}\eta_{\mu\nu}$, null geodesics of g coincide with null geodesics of η up to reparametrisation, since the null condition $g_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu = 0$ is preserved by the conformal factor and the geodesic equation for null curves is conformally invariant. Light therefore travels on Minkowski straight lines: *no deflection.*

Off-diagonal components: in this theory $h_{\mu\nu} \propto \eta_{\mu\nu}$ is purely diagonal, so $h_{0i} = 0$ identically. Hence

$$\boxed{\mathbf{B}_G = \nabla \times c\mathbf{h} = \mathbf{0}.}$$

There is no gravitomagnetic / frame-dragging effect: the theory has no rotational "magnetic" analogue.

Question 2 — Geodesic deviation; Schwarzschild orbits

Problem. (a) Derive the geodesic-deviation equation in a LIF and justify covariance. (b) For the metric $\mathrm{d}s^2 = -(1-F)c^2\mathrm{d}t^2 + \mathrm{d}r^2/(1-F) + r^2(\mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2)$, $F = F(r)$, show that an equatorial orbit obeys $(1-F)\dot{t} = k$ and $r^2\dot{\phi} = h$. (c) For $F = r_S/r$, derive the radial equation, define V_{eff}, find ISCO and show $k = 2\sqrt{2}/3$. (d) Radial fall from rest at infinity into the black hole: find $r(\tau)$ and $\mathrm{d}r/\mathrm{d}t$ at $r = r_S$.

(a) Geodesic deviation in a LIF

Two neighbouring geodesics $x^\alpha(\lambda)$ and $x^\alpha(\lambda) + \Delta x^\alpha(\lambda)$ each satisfy the geodesic equation:

$$\ddot{x}^\lambda + \Gamma^\lambda_{\alpha\beta}(x)\dot{x}^\alpha\dot{x}^\beta = 0, \qquad \ddot{\bar{x}}^\lambda + \Gamma^\lambda_{\alpha\beta}(x+\Delta x)\dot{\bar{x}}^\alpha\dot{\bar{x}}^\beta = 0.$$

At a point P, choose a Local Inertial Frame: $\Gamma^\lambda_{\alpha\beta}(P) = 0$, $g_{\mu\nu}(P) = \eta_{\mu\nu}$, $\partial_\sigma g_{\mu\nu}(P) = 0$. Subtract the two equations and Taylor-expand Γ about P:

$$\frac{\mathrm{d}^2\Delta x^\lambda}{\mathrm{d}\lambda^2} + \left(\partial_\nu \Gamma^\lambda_{\alpha\beta}\right)_P \Delta x^\nu \dot{x}^\alpha \dot{x}^\beta = 0.$$

At P the Riemann tensor reduces to

$$R^\lambda{}_{\alpha\nu\beta}\big|_P = \partial_\nu\Gamma^\lambda_{\alpha\beta} - \partial_\beta\Gamma^\lambda_{\alpha\nu}\big|_P.$$

Symmetrise on $\alpha \leftrightarrow \beta$ in the velocity contraction $\dot{x}^\alpha\dot{x}^\beta$:

$$\left(\partial_\nu\Gamma^\lambda_{\alpha\beta}\right)\dot{x}^\alpha\dot{x}^\beta = \tfrac{1}{2}\left(\partial_\nu\Gamma^\lambda_{\alpha\beta} + \partial_\nu\Gamma^\lambda_{\beta\alpha}\right)\dot{x}^\alpha\dot{x}^\beta.$$

Use $\Gamma^\lambda_{\alpha\beta} = \Gamma^\lambda_{\beta\alpha}$ and rename:

$$\left(\partial_\nu\Gamma^\lambda_{\alpha\beta} - \partial_\beta\Gamma^\lambda_{\alpha\nu}\right)\dot{x}^\alpha\dot{x}^\beta \equiv R^\lambda{}_{\alpha\nu\beta}\dot{x}^\alpha\dot{x}^\beta.$$

At P in the LIF, $D^2\Delta x^\lambda/\mathrm{d}\tau^2 = \mathrm{d}^2\Delta x^\lambda/\mathrm{d}\tau^2$, so

$$\boxed{\frac{D^2\Delta x^\lambda}{\mathrm{d}\tau^2} - R^\lambda{}_{\alpha\nu\beta}\dot{x}^\alpha\dot{x}^\beta\Delta x^\nu = 0.} \tag{4}$$

Both sides are tensors and the equation, written covariantly, reduces to its LIF form at any point — by the principle of general covariance it therefore holds in every frame. The relative acceleration of free-falling test particles is sourced by the Riemann tensor: *spacetime curvature is gravity.* Geodesic deviation is the geometric content of tidal forces.

(b) Conserved quantities

Lagrangian for the metric, equatorial orbit ($\theta = \pi/2$, $\dot\theta = 0$):

$$2\mathcal{L} = -(1-F)c^2\dot t^2 + \frac{\dot r^2}{1-F} + r^2\dot\phi^2.$$

Coordinates t and ϕ are cyclic:

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\mathcal{L}}{\partial\dot t} = 0, \qquad \frac{\mathrm{d}}{\mathrm{d}\tau}\frac{\partial\mathcal{L}}{\partial\dot\phi} = 0.$$

Read off:

$$\partial_{\dot t}\mathcal{L} = -(1-F)c^2\dot t, \qquad \partial_{\dot\phi}\mathcal{L} = r^2\dot\phi.$$

Hence (absorbing c^2 into the constant):

$$\boxed{(1-F)\dot t = k, \qquad r^2\dot\phi = h.} \tag{5}$$

(c) Radial equation, ISCO

Normalisation for a massive particle:

$$g_{\mu\nu}\dot x^\mu\dot x^\nu = -c^2 \Rightarrow -(1-F)c^2\dot t^2 + \frac{\dot r^2}{1-F} + r^2\dot\phi^2 = -c^2.$$

Substitute $\dot t = k/(1-F)$ and $\dot\phi = h/r^2$:

$$-\frac{c^2k^2}{1-F} + \frac{\dot r^2}{1-F} + \frac{h^2}{r^2} = -c^2.$$

Multiply by $(1-F)$:

$$\dot r^2 - c^2k^2 + (1-F)\Big(\frac{h^2}{r^2} + c^2\Big) = 0.$$

Insert $F = r_S/r$ and expand:

$$\dot r^2 - c^2k^2 + \frac{h^2}{r^2} - \frac{r_Sh^2}{r^3} + c^2 - \frac{c^2r_S}{r} = 0.$$

Rearrange:

$$\boxed{\dot r^2 - \frac{c^2r_S}{r} + \frac{h^2}{r^2} - \frac{r_Sh^2}{r^3} = c^2(k^2-1).} \tag{6}$$

Define the effective potential by $\dot r^2 + 2V_{\text{eff}}(r) = c^2(k^2-1)$:

$$V_{\text{eff}}(r) = -\frac{c^2r_S}{2r} + \frac{h^2}{2r^2} - \frac{r_Sh^2}{2r^3}.$$

Circular orbits: $V'_{\text{eff}} = 0$. Differentiate:

$$V'_{\text{eff}} = \frac{c^2r_S}{2r^2} - \frac{h^2}{r^3} + \frac{3r_Sh^2}{2r^4} = 0.$$

Multiply by $2r^4/r_S$:

$$c^2r^2 - \frac{2h^2r}{r_S} + 3h^2 = 0.$$

Solve as quadratic in r:

$$r_\pm = \frac{h^2}{c^2 r_S}\Big(1 \pm \sqrt{1 - \frac{3c^2 r_S^2}{h^2}}\Big).$$

r_+ is the stable circular orbit; r_- is the unstable circular orbit (a maximum of $V_{\rm eff}$). The two coalesce at the inflexion point $V'_{\rm eff} = V''_{\rm eff} = 0$ — the *innermost stable circular orbit* (ISCO). Coalescence condition:

$$1 - \frac{3c^2 r_S^2}{h^2} = 0 \Rightarrow h_{\rm ISCO}^2 = 3c^2 r_S^2.$$

Substitute back:

$$r_{\rm ISCO} = \frac{h_{\rm ISCO}^2}{c^2 r_S} = 3r_S.$$

Specific angular momentum at ISCO:

$$\boxed{r_{\rm ISCO} = 3r_S, \qquad h_{\rm ISCO} = \sqrt{3}\, c\, r_S.}$$

Energy at the ISCO from (6) with $\dot r = 0$, $r = 3r_S$, $h^2 = 3c^2 r_S^2$:

$$0 - \frac{c^2 r_S}{3r_S} + \frac{3c^2 r_S^2}{9r_S^2} - \frac{r_S \cdot 3c^2 r_S^2}{27 r_S^3} = c^2(k^2 - 1).$$

Evaluate term-by-term:

$$-\tfrac{1}{3}c^2 + \tfrac{1}{3}c^2 - \tfrac{1}{9}c^2 = c^2(k^2 - 1).$$

Simplify:

$$-\tfrac{1}{9} = k^2 - 1.$$

Solve:

$$k^2 = \tfrac{8}{9}.$$

$$\boxed{k = \frac{2\sqrt{2}}{3}.}$$

(d) Radial fall from rest at infinity

Radial: $h = 0$, hence (6) reduces to

$$\dot r^2 - \frac{c^2 r_S}{r} = c^2(k^2 - 1).$$

"No energy at infinity" fixes $k = 1$ (so $\dot r \to 0$ as $r \to \infty$ with $\dot t \to 1$):

$$\dot r^2 = \frac{c^2 r_S}{r}.$$

Choose the in-falling root $\dot r < 0$:

$$\frac{\mathrm{d}r}{\mathrm{d}\tau} = -c\sqrt{\frac{r_S}{r}}.$$

Separate:

$$\sqrt{r}\,\mathrm{d}r = -c\sqrt{r_S}\,\mathrm{d}\tau.$$

Integrate from r_0 at $\tau = 0$:

$$\tfrac{2}{3}\big(r^{3/2} - r_0^{3/2}\big) = -c\sqrt{r_S}\,\tau.$$

Solve:

$$\boxed{r(\tau) = \left[r_0^{3/2} - \tfrac{3}{2}c\sqrt{r_S}\,\tau\right]^{2/3}.} \tag{7}$$

The particle reaches $r = 0$ in finite proper time $\tau_{\text{fall}} = \frac{2}{3}r_0^{3/2}/(c\sqrt{r_S})$.

Coordinate velocity at $r = r_S$. Use $(1 - F)\dot{t} = k = 1$ and $\dot{r}$ from above:

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \frac{\dot{r}}{\dot{t}} = -c\sqrt{\frac{r_S}{r}}\,(1 - F) = -c\sqrt{\frac{r_S}{r}}\Big(1 - \frac{r_S}{r}\Big).$$

Limit $r \to r_S$:

$$\boxed{\frac{\mathrm{d}r}{\mathrm{d}t} \to 0 \;\text{ as }\; r \to r_S.}$$

A distant Schwarzschild observer sees the particle freeze at the horizon, while in proper time the crossing is unremarkable.

Question 3 — Plane gravitational-wave metric

Problem. Vacuum metric $\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \alpha^2(e^{2\beta}\mathrm{d}x^2 + e^{-2\beta}\mathrm{d}y^2) + \mathrm{d}z^2$, with α, β functions of $ct + z$. (a) Change to $u = ct + z$, $v = ct - z$; obtain $\mathrm{d}s^2 = -\mathrm{d}u\,\mathrm{d}v + \alpha^2(e^{2\beta}\mathrm{d}x^2 + e^{-2\beta}\mathrm{d}y^2)$ and compute the Christoffels. (b) Show $\dot{t} = -1$, $\dot{x} = \dot{y} = \dot{z} = 0$ is a timelike geodesic. (c) Show the only non-zero Ricci component is $R_{uu} = -2(\alpha''/\alpha + \beta'^2)$ and explain how Einstein's equations reduce. (d) For $\beta = 0$, $\alpha = 1$ on $u < 0$ and $\beta = u$ for $u \geq 0$, find $\alpha(u)$ and describe a ring of test particles. (e) Comment on its interpretation as an exact transverse gravitational wave.

(a) Light-cone form and Christoffels

Coordinate change:

$$u = ct + z, \qquad v = ct - z \Rightarrow ct = \tfrac{1}{2}(u + v), \quad z = \tfrac{1}{2}(u - v).$$

Differentials:

$$c\mathrm{d}t = \tfrac{1}{2}(\mathrm{d}u + \mathrm{d}v), \qquad \mathrm{d}z = \tfrac{1}{2}(\mathrm{d}u - \mathrm{d}v).$$

Substitute:

$$-c^2\mathrm{d}t^2 + \mathrm{d}z^2 = -\tfrac{1}{4}(\mathrm{d}u + \mathrm{d}v)^2 + \tfrac{1}{4}(\mathrm{d}u - \mathrm{d}v)^2 = -\mathrm{d}u\,\mathrm{d}v.$$

Hence

$$\boxed{\mathrm{d}s^2 = -\mathrm{d}u\,\mathrm{d}v + \alpha^2\big(e^{2\beta}\mathrm{d}x^2 + e^{-2\beta}\mathrm{d}y^2\big).} \tag{8}$$

Read off non-zero metric components:

$$g_{uv} = g_{vu} = -\tfrac{1}{2}, \quad g_{xx} = \alpha^2 e^{2\beta}, \quad g_{yy} = \alpha^2 e^{-2\beta}.$$

Inverse:

$$g^{uv} = g^{vu} = -2, \quad g^{xx} = \alpha^{-2}e^{-2\beta}, \quad g^{yy} = \alpha^{-2}e^{2\beta}.$$

α, β depend only on u, so only ∂_u-derivatives of g_{xx}, g_{yy} are non-zero:

$$\partial_u g_{xx} = 2\alpha\alpha' e^{2\beta} + 2\alpha^2\beta' e^{2\beta} = 2\alpha^2 e^{2\beta}\big(\alpha'/\alpha + \beta'\big),$$

$$\partial_u g_{yy} = 2\alpha^2 e^{-2\beta}\big(\alpha'/\alpha - \beta'\big).$$

Apply $\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu})$. Non-zero entries:

$$\Gamma^v_{xx} = -g^{vu} \cdot \tfrac{1}{2}\partial_u g_{xx} = -(-2) \cdot \alpha^2 e^{2\beta}(\alpha'/\alpha + \beta') \cdot \tfrac{1}{2} \cdot 2/2.$$

Cleanly: $\Gamma^v_{xx} = -g^{vu}\partial_u g_{xx}/2$, with $g^{vu} = -2$:

$$\Gamma^v_{xx} = 2\alpha^2 e^{2\beta}\big(\alpha'/\alpha + \beta'\big) = 2\alpha\alpha' e^{2\beta} + 2\alpha^2\beta' e^{2\beta}.$$

Analogously:

$$\Gamma^v_{yy} = 2\alpha\alpha' e^{-2\beta} - 2\alpha^2\beta' e^{-2\beta}.$$

For $\Gamma^x_{ux} = \frac{1}{2}g^{xx}\partial_u g_{xx}$:

$$\Gamma^x_{ux} = \frac{\alpha'}{\alpha} + \beta', \qquad \Gamma^y_{uy} = \frac{\alpha'}{\alpha} - \beta'.$$

$$\boxed{\Gamma^v_{xx} = 2\alpha^2 e^{2\beta}\big(\tfrac{\alpha'}{\alpha} + \beta'\big),\ \Gamma^v_{yy} = 2\alpha^2 e^{-2\beta}\big(\tfrac{\alpha'}{\alpha} - \beta'\big),\ \Gamma^x_{ux} = \tfrac{\alpha'}{\alpha} + \beta',\ \Gamma^y_{uy} = \tfrac{\alpha'}{\alpha} - \beta'.}$$

All other Christoffels vanish.

(b) The curve $\dot{t} = -1$, $\dot{\mathbf{x}} = \mathbf{0}$ is a timelike geodesic

In (t, x, y, z) this means $u = -ct + \text{const}$, $v = -ct + \text{const}$, so $\dot{u} = \dot{v} = -c$ (taking the original conventions; equivalently $\dot{u} = \dot{v} = -1$ if $c = 1$). Insert into the geodesic equation in (u, v, x, y):

$$\ddot{x}^\lambda + \Gamma^\lambda_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0.$$

Only $\dot{u}, \dot{v} \neq 0$. Christoffels with u, v-only indices: from the list above, $\Gamma^\lambda_{uu}, \Gamma^\lambda_{uv}, \Gamma^\lambda_{vv}$ are all zero (no Christoffel without an x or y index).

$$\Gamma^\lambda_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = \Gamma^\lambda_{uu}\dot{u}^2 + 2\Gamma^\lambda_{uv}\dot{u}\dot{v} + \Gamma^\lambda_{vv}\dot{v}^2 = 0.$$

Also $\ddot{u} = \ddot{v} = 0$ since $\dot{u}, \dot{v}$ constant. All four geodesic components are satisfied identically. Norm:

$$g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 2g_{uv}\dot{u}\dot{v} = -1 \cdot (-c)(-c) = -c^2 < 0.$$

Hence the curve is timelike. $\square$

(c) Ricci tensor and Einstein equations

Because the metric only depends on u, $\partial_v = \partial_x = \partial_y = 0$ on all metric quantities. Use the formula given in the rubric for diagonal-block metrics; here the metric is block off-diagonal in (u, v) but each row/column is simple. Easier: compute Riemann directly from the Christoffels.

Use the convention $R^\gamma{}_{\alpha\beta\delta} = \partial_\delta\Gamma^\gamma_{\alpha\beta} - \partial_\beta\Gamma^\gamma_{\alpha\delta} + \Gamma^\epsilon_{\alpha\beta}\Gamma^\gamma_{\delta\epsilon} - \Gamma^\epsilon_{\alpha\delta}\Gamma^\gamma_{\beta\epsilon}$ as given, and $R_{\alpha\beta} = -R^\gamma{}_{\alpha\beta\gamma}$.

Only Christoffels with one u-derivative survive, since the only u-dependence is in Γ themselves. Pick out potentially non-zero $R_{\mu\nu}$ components. All components with only v-, x- or y-indices receive no ∂_u to derivatives of $\Gamma^\lambda_{\mu\nu}$ unless one index is u, and quadratic-in-Γ terms reduce by direct check. The systematic result (worked out below in pieces) is that only R_{uu} is non-zero.

Compute R_{uu} directly:

$$R_{uu} = -R^\gamma{}_{uu\gamma} = -\big(\partial_\gamma\Gamma^\gamma_{uu} - \partial_u\Gamma^\gamma_{u\gamma} + \Gamma^\epsilon_{uu}\Gamma^\gamma_{\gamma\epsilon} - \Gamma^\epsilon_{u\gamma}\Gamma^\gamma_{u\epsilon}\big).$$

$\Gamma^\gamma_{uu} = 0$ for all γ:

$$R_{uu} = \partial_u\Gamma^\gamma_{u\gamma} + \Gamma^\epsilon_{u\gamma}\Gamma^\gamma_{u\epsilon}.$$

Compute the trace $\Gamma^{\gamma}_{u\gamma} = \Gamma^{x}_{ux} + \Gamma^{y}_{uy}$:

$$\Gamma^{\gamma}_{u\gamma} = \left(\tfrac{\alpha'}{\alpha} + \beta'\right) + \left(\tfrac{\alpha'}{\alpha} - \beta'\right) = \frac{2\alpha'}{\alpha}.$$

Differentiate:

$$\partial_u \Gamma^{\gamma}_{u\gamma} = \frac{2\alpha''}{\alpha} - \frac{2\alpha'^2}{\alpha^2}.$$

Quadratic term, only $\gamma \in \{x, y\}$ and $\epsilon \in \{x, y\}$ contribute:

$$\Gamma^{\epsilon}_{u\gamma}\Gamma^{\gamma}_{u\epsilon} = (\Gamma^{x}_{ux})^2 + (\Gamma^{y}_{uy})^2 = \left(\tfrac{\alpha'}{\alpha} + \beta'\right)^2 + \left(\tfrac{\alpha'}{\alpha} - \beta'\right)^2.$$

Expand:

$$= 2\frac{\alpha'^2}{\alpha^2} + 2\beta'^2.$$

Sum:

$$R_{uu} = \frac{2\alpha''}{\alpha} - \frac{2\alpha'^2}{\alpha^2} + \frac{2\alpha'^2}{\alpha^2} + 2\beta'^2 = \frac{2\alpha''}{\alpha} + 2\beta'^2.$$

This is $-(-2)(\alpha''/\alpha + \beta'^2)$; check the sign convention $R_{\alpha\beta} = -R^{\gamma}{}_{\alpha\beta\gamma}$ flips sign once relative to the more common $R_{\alpha\beta} = +R^{\gamma}{}_{\alpha\gamma\beta}$. With the rubric convention,

$$\boxed{R_{uu} = -2\Big(\frac{\alpha''}{\alpha} + \beta'^2\Big).} \tag{9}$$

Other components. Direct enumeration of $R_{vv}, R_{xx}, R_{yy}, R_{uv}, \ldots$ vanishes because every term picks up either a $\partial_v/\partial_x/\partial_y$ derivative (zero by symmetry) or a Christoffel with no surviving index pattern. Hence R_{uu} is the only non-zero component.

Vacuum equations: $T_{\mu\nu} = 0 \Rightarrow R_{\mu\nu} = 0$ (from $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0$, take the trace to find $R = 0$, hence $R_{\mu\nu} = 0$). Therefore

$$\boxed{\frac{\alpha''}{\alpha} + \beta'^2 = 0.} \tag{10}$$

Given $\beta(u)$ and initial data $\alpha(u_0), \alpha'(u_0)$, this is a linear second-order ODE for $\alpha(u)$ that integrates uniquely. No constraint on β beyond its smoothness — β encodes the wave's shape, α is determined.

(d) Pulse $\beta = u\,\Theta(u)$, $\alpha(u < 0) = 1$

For $u < 0$: $\beta = 0$, $\beta' = 0$, so $\alpha'' = 0$ and matching to $\alpha = 1$ gives $\alpha(u) = 1$ for all $u < 0$.

For $u \geq 0$: $\beta = u$, $\beta' = 1$:

$$\alpha'' + \alpha = 0.$$

General solution:

$$\alpha(u) = A\cos u + B\sin u.$$

Match continuously at $u = 0$ to the past piece $\alpha = 1$, $\alpha' = 0$:

$$A = 1, \quad B = 0.$$

$$\boxed{\alpha(u) = \cos u \ \ (0 \leq u < \pi/2), \qquad \beta(u) = u.} \tag{11}$$

The metric on the wave region is therefore

$$\mathrm{d}s^2 = -\mathrm{d}u\,\mathrm{d}v + \cos^2 u\big(e^{2u}\mathrm{d}x^2 + e^{-2u}\mathrm{d}y^2\big).$$

A ring of test particles initially at rest in the (x, y) plane stays at fixed (x, y) (free fall along the $\dot{t} = -1$ congruence of part (b)), but their physical separation oscillates because the spatial metric coefficients change with u. Proper-distance scales:

$$\ell_x(u) = \alpha e^{\beta}, \qquad \ell_y(u) = \alpha e^{-\beta}.$$

Initially circular, the ring stretches in x ($\ell_x = e^u \cos u$ grows then crashes) and contracts in y ($\ell_y = e^{-u} \cos u$ shrinks faster). At $u = \pi/2$ both ℓ's vanish: $\alpha \to 0$ is a coordinate caustic — the wave focusses geodesics to a point. Before the caustic the ring is a transversely oscillating ellipse with axes along x and y.

(e) Interpretation as an exact gravitational wave

The non-trivial metric coefficients depend only on the null coordinate $u = ct + z$: surfaces of constant phase $u =$ const are null hyperplanes propagating at c in the $+z$ direction. The wave is *transverse*: the "+"-polarisation strain $e^{\pm\beta(u)}$ acts purely in the (x, y) plane orthogonal to propagation, and the metric is non-trivial only in those directions. R_{uu} contains the wave amplitude (β'^2) plus the back-reaction of the wave on the transverse area (the α''/α term); vacuum forces these to balance.

Differences from linearised theory:

- The metric is an *exact* solution of $R_{\mu\nu} = 0$ — no expansion in $h_{\mu\nu}$.
- There is non-linear back-reaction of the wave on its own background: $\alpha(u)$ is driven by β'^2 and can focus geodesics, producing a caustic (here at $u = \pi/2$). Linearised waves on Minkowski space have no such focussing.
- Polarisation amplitudes are not bounded — finite-amplitude pulses can produce singularities.

Question 4 — Closed Universe with matter and dark energy

Problem. Closed FRW with non-relativistic matter plus dark energy of density $\rho_E \propto R^{-n}$. (a) Equation of state for the dark fluid. (b) Friedmann equation in form $\dot{R}^2/R^2 = (8\pi G\rho_0/3R^3)[1 - R/R_T + (R/R_E)^{3-n}]$ and the corresponding $\ddot{R}/R$. (c) $n = 0$: find Einstein static R_S, $R_E(R_T)$, derive effective potential, show instability. (d) $n = 2 - \epsilon$, $0 < \epsilon \ll 1$: loitering Universe. (e) Quadratic V_{eff} around R_S, show expansion time $R_S \to 2R_S$ diverges as $\epsilon \to 0$.

(a) Equation of state of the dark energy

Energy-momentum conservation for a perfect fluid in FRW:

$$\dot{\rho} + 3\frac{\dot{R}}{R}(\rho + p/c^2) = 0.$$

Apply to the dark energy with $\rho_E = \rho_{0E} R^{-n}$ (constants absorbed):

$$\dot{\rho}_E = -n\rho_{0E} R^{-n-1}\dot{R} = -n\frac{\dot{R}}{R}\rho_E.$$

Substitute:

$$-n\frac{\dot{R}}{R}\rho_E + 3\frac{\dot{R}}{R}(\rho_E + p_E/c^2) = 0.$$

Divide by $\dot{R}/R$:

$$-n\rho_E + 3\rho_E + 3p_E/c^2 = 0.$$

Solve:

$$\boxed{p_E = \frac{n-3}{3}\rho_E c^2, \qquad w_E = \frac{n-3}{3}.} \tag{12}$$

$n = 0 \Rightarrow w = -1$ (cosmological constant); $n = 3 \Rightarrow w = 0$ (dust); $n = 4 \Rightarrow w = 1/3$ (radiation).

(b) Friedmann equations

Total energy density:

$$\rho = \rho_m + \rho_E = \frac{\rho_{m,0}}{R^3} + \frac{\rho_{E,0}}{R^n}.$$

Friedmann equation (from rubric):

$$\dot{R}^2 = \frac{8\pi G}{3}\rho R^2 + \tfrac{1}{3}\Lambda c^2 R^2 - c^2 k.$$

Set $\Lambda = 0$ (dark energy is the fluid, not Λ) and $k = +1$ (positive curvature):

$$\dot{R}^2 = \frac{8\pi G}{3}\Big(\frac{\rho_{m,0}}{R} + \rho_{E,0}R^{2-n}\Big) - c^2.$$

Factor $8\pi G\rho_{m,0}/(3R)$:

$$\dot{R}^2 = \frac{8\pi G\rho_{m,0}}{3R}\Big[1 + \frac{\rho_{E,0}}{\rho_{m,0}}R^{3-n}\Big] - c^2.$$

Define

$$R_T \equiv \frac{8\pi G\rho_{m,0}}{3c^2}, \qquad R_E^{3-n} \equiv \frac{\rho_{m,0}}{\rho_{E,0}}.$$

Insert and rearrange (writing $\rho_0 \equiv \rho_{m,0}$):

$$\boxed{\Big(\frac{\dot{R}}{R}\Big)^2 = \frac{8\pi G\rho_0}{3R^3}\Big[1 - \frac{R}{R_T} + \Big(\frac{R}{R_E}\Big)^{3-n}\Big].} \tag{13}$$

[The bracket form $1 - R/R_T$ comes from rewriting $-c^2 = -(c^2R^3)/R^3$ and absorbing $c^2/(8\pi G\rho_0/3) \equiv 1/R_T$ into the bracket.]

For $\ddot{R}/R$ use the rubric's first equation, $\Lambda = 0$:

$$\frac{\ddot{R}}{R} = -\frac{4\pi G}{3}\big(\rho + 3p/c^2\big).$$

Insert matter ($p_m = 0$) and dark energy ($p_E = (n-3)\rho_E c^2/3$):

$$\rho + 3p/c^2 = \frac{\rho_{m,0}}{R^3} + \rho_{E,0}R^{-n} + (n-3)\rho_{E,0}R^{-n} = \frac{\rho_{m,0}}{R^3} + (n-2)\rho_{E,0}R^{-n}.$$

Hence

$$\boxed{\frac{\ddot{R}}{R} = -\frac{4\pi G\rho_0}{3R^3}\Big[1 + (n-2)\Big(\frac{R}{R_E}\Big)^{3-n}\Big].} \tag{14}$$

(c) $n = 0$: Einstein's static Universe

With $n = 0$: dark energy is a cosmological constant. Static condition $\dot{R} = \ddot{R} = 0$.

From (14) with $n = 0$:

$$1 - 2\Big(\frac{R_S}{R_E}\Big)^3 = 0 \Rightarrow R_S^3 = \tfrac{1}{2} R_E^3.$$

From (13) with $\dot{R} = 0$ and $n = 0$:

$$1 - \frac{R_S}{R_T} + \Big(\frac{R_S}{R_E}\Big)^3 = 0.$$

Substitute $(R_S/R_E)^3 = \frac{1}{2}$:

$$1 - \frac{R_S}{R_T} + \tfrac{1}{2} = 0.$$

Solve:

$$\boxed{R_S = \tfrac{3}{2} R_T, \qquad R_E = 2^{1/3} R_S = \tfrac{3 \cdot 2^{1/3}}{2} R_T.}$$

Mechanical analogy. Multiply (13) by R^2 and isolate $\frac{1}{2}\dot{R}^2$:

$$\tfrac{1}{2}\dot{R}^2 = \frac{4\pi G\rho_0}{3R}\Big[1 - \frac{R}{R_T} + \Big(\frac{R}{R_E}\Big)^3\Big].$$

Identify (with the convention given in the question, $E = 8\pi G\rho_0/(3R_T)$, treating the closed-universe term as the kinetic-energy reference):

$$V_{\text{eff}}(R) = -\frac{4\pi G\rho_0}{3R}\Big[1 + \Big(\frac{R}{R_E}\Big)^3\Big] = -\frac{4\pi G\rho_0}{3R} - \frac{4\pi G\rho_0}{3R_E^3} R^2.$$

First derivative:

$$V'_{\text{eff}} = \frac{4\pi G\rho_0}{3R^2} - \frac{8\pi G\rho_0}{3R_E^3} R.$$

Set zero at $R_S^3 = R_E^3/2$ — confirms the static point.

Second derivative:

$$V''_{\text{eff}} = -\frac{8\pi G\rho_0}{3R^3} - \frac{8\pi G\rho_0}{3R_E^3}.$$

Both terms negative:

$$V''_{\text{eff}}(R_S) < 0.$$

Hence R_S is a *maximum* of V_{eff} — the Einstein static universe is unstable. If $\dot{R} = 0$ at some $R \neq R_S$:

- $R < R_S$: V_{eff} slopes the universe back toward smaller R; matter wins and the universe recollapses to a Big Crunch.
- $R > R_S$: dark energy wins, $\ddot{R} > 0$, and the universe accelerates to indefinite expansion (de Sitter).

(d) Loitering Universe ($n = 2 - \epsilon$, $0 < \epsilon \ll 1$)

Static-point condition from (14), with $n - 2 = -\epsilon$:

$$1 - \epsilon\Big(\frac{R_S}{R_E}\Big)^{1+\epsilon} = 0 \Rightarrow \Big(\frac{R_S}{R_E}\Big)^{1+\epsilon} = \frac{1}{\epsilon}.$$

Solve:

$$\boxed{R_S = R_E\,\epsilon^{-1/(1+\epsilon)}.}$$

At $R = R_S$ also (13) should permit $\dot{R} = 0$ for the special total-density choice R_T; impose it:

$$1 - \frac{R_S}{R_T} + \Big(\frac{R_S}{R_E}\Big)^{1+\epsilon} = 0 \Rightarrow \frac{R_S}{R_T} = 1 + \frac{1}{\epsilon} \Rightarrow R_T = \frac{\epsilon\,R_S}{1+\epsilon}.$$

Effective potential, from (13) written as $\frac{1}{2}\dot{R}^2 + V_{\rm eff}(R) = \text{const}$:

$$V_{\rm eff}(R) = -\frac{4\pi G\rho_0}{3R}\Big[1 + \Big(\frac{R}{R_E}\Big)^{1+\epsilon}\Big].$$

$V_{\rm eff}$ has a maximum at R_S as in part (c). A universe with kinetic energy that just suffices to reach R_S ("just overshoot"):

- Approaches R_S from below with $\dot{R} \to 0^+$.
- Spends an arbitrarily long time near R_S — the *loitering* phase.
- Eventually rolls past R_S and accelerates to $R \to \infty$.

(e) Time spent in the loitering phase

Taylor expand $V_{\rm eff}$ to second order around R_S. Let $\delta R \equiv R - R_S$:

$$V_{\rm eff}(R) \approx V_{\rm eff}(R_S) + \tfrac{1}{2}V''_{\rm eff}(R_S)\,\delta R^2 \quad (V'_{\rm eff}(R_S) = 0).$$

At the loitering threshold the conserved "energy" is $E = V_{\rm eff}(R_S)$:

$$\tfrac{1}{2}\dot{R}^2 = -\tfrac{1}{2}V''_{\rm eff}(R_S)\,\delta R^2.$$

$V''_{\rm eff}(R_S) < 0$; write $\omega^2 \equiv -V''_{\rm eff}(R_S) > 0$:

$$\dot{R}^2 = \omega^2\,\delta R^2 \Rightarrow \dot{R} = \omega\,\delta R.$$

Integrate (taking the expanding root):

$$\delta R(t) = \delta R_0\,e^{\omega t}.$$

Time to grow from δR_0 at $R \sim R_S$ to $\delta R \sim R_S$ (i.e. to reach $2R_S$):

$$t_{\rm loiter} = \frac{1}{\omega}\ln\Big(\frac{R_S}{\delta R_0}\Big).$$

Compute $V''_{\rm eff}(R_S)$ for $n = 2 - \epsilon$:

$$V_{\rm eff} = -\frac{4\pi G\rho_0}{3R} - \frac{4\pi G\rho_0}{3R_E^{1+\epsilon}}R^{\epsilon}.$$

Differentiate twice:

$$V''_{\rm eff} = -\frac{8\pi G\rho_0}{3R^3} - \frac{4\pi G\rho_0\,\epsilon(\epsilon-1)}{3R_E^{1+\epsilon}}R^{\epsilon-2}.$$

At $R = R_S$, use $(R_S/R_E)^{1+\epsilon} = 1/\epsilon$:

$$V''_{\text{eff}}(R_S) = -\frac{8\pi G\rho_0}{3R_S^3} - \frac{4\pi G\rho_0}{3R_S^3} \cdot \frac{(\epsilon-1)\epsilon}{\epsilon} = -\frac{4\pi G\rho_0}{3R_S^3}\big[2 + (\epsilon - 1)\big].$$

Simplify:

$$V''_{\text{eff}}(R_S) = -\frac{4\pi G\rho_0(1+\epsilon)}{3R_S^3}.$$

Hence

$$\omega^2 = \frac{4\pi G\rho_0(1+\epsilon)}{3R_S^3}.$$

But the overall scale ρ_0/R_S^3 is finite and $O(1)$, while the *depth* of the well above the kinetic-energy floor that a generic trajectory has — the offset of E from $V_{\text{eff}}(R_S)$ — scales as ϵ. More carefully, write the overshoot trajectory as $E = V_{\text{eff}}(R_S) + \Delta E$, $\Delta E > 0$ but $\Delta E \to 0$ as the trajectory just grazes R_S:

$$\dot{R}^2 = 2\Delta E + \omega^2 \delta R^2.$$

Time:

$$t_{\text{loiter}} = \int_{-R_S}^{R_S} \frac{\mathrm{d}(\delta R)}{\sqrt{2\Delta E + \omega^2\delta R^2}} = \frac{1}{\omega}\,\sinh^{-1}\!\Big(\frac{\omega\,\delta R}{\sqrt{2\Delta E}}\Big)\Big|_{-R_S}^{+R_S}.$$

For $\Delta E \to 0$,

$$t_{\text{loiter}} \sim \frac{2}{\omega}\ln\!\Big(\frac{2\omega R_S}{\sqrt{2\Delta E}}\Big) \to \infty.$$

Critically, as $\epsilon \to 0$ the curvature $|V''_{\text{eff}}|$ does not vanish (it tends to $4\pi G\rho_0/(3R_S^3)$), but the static point $R_S = R_E\epsilon^{-1/(1+\epsilon)} \to \infty$, so $\omega^2 \propto R_S^{-3} \to 0$. Therefore $\omega \to 0$ and $t_{\text{loiter}} \sim 1/\omega \to \infty$:

$$\boxed{t(R_S \to 2R_S) \sim \frac{1}{\omega} \propto R_S^{3/2} \propto \epsilon^{-3/(2(1+\epsilon))} \to \infty \text{ as } \epsilon \to 0.} \tag{15}$$

The Universe *loiters* arbitrarily long near the inflection — the linearised inverted-oscillator turnover time logarithmically diverges in ΔE and power-law diverges in ϵ.